Grounding the Loop on Both Sides: Interaction as a Third Test-Time Compute Axis, and Why Its Gains Are Invisible Without Grounded Evaluation

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Abstract

Test-time compute is dominated by two scaling axes that both operate inside a model's own token space: *reasoning* (longer chains of thought) and *sampling* (best-of-N). Both are bounded, by the data-processing inequality (DPI), by the information already in the weights and prompt. A third axis—*interaction* with an environment that returns grounded feedback—can exceed that bound by importing information from outside the model. We argue that a single variable, **grounding**, governs this axis, and that it must hold on *both* sides of the loop: grounded *feedback* drives improvement past the reasoning/sampling ceiling, and grounded *evaluation* is required even to *measure* the improvement. We formalize the first claim as a data-processing-inequality argument over a Type 0–3 feedback taxonomy. For the second, we show that on rendered visual artifacts the default vision–language-model (VLM) judge is structurally blind to the geometric defects the loop fixes—it rates 14 of 15 broken academic figures "perfect"—and that a VLM *reviewer* even makes layouts worse. Replacing it with a deterministic DOM-geometry instrument (overlap, clipping, overflow, and box-group misalignment, measured exactly from the rendered layout) reveals large, significant defect reductions under grounded feedback across **four** visual modalities measured identically: academic figures −74%, dense slides −73%, responsive web pages −47%, and SVG/CSS animations −40% (all paired sign-test $p < 0.002$; all bootstrap 95% CIs exclude zero). On execution-grounded code the harness lifts a frontier model from 66.7% to a strict 100% pass-rate, and an in-modality control isolates the *execution* of tests—not the extra reviewing pass—as the cause; this replicates across three model families and a held-out suite, and we extend the grounded-geometry harness to a second model family (Gemini 3.1 Pro) scored by the same model-free instrument. At a fixed 20K-token budget, reasoning-only and best-of-N saturate (73.3%/86.7%) while every interaction strategy reaches $\geq 97.8\%$, and the proposer–reviewer harness does so at the lowest token cost with zero seed variance. Finally, the harness's behavior partially distills into an 8B student. Interaction scaling is real, distinct, and predictable—but only visible when both the feedback and the metric are grounded.

1 Introduction

Scaling test-time compute has become the dominant lever for improving large models on artifact-producing tasks—code, web pages, slides, figures, animations, video edits, research reports. The literature has formalized two axes, and both operate *inside the model's own token space. Reasoning* *scaling* [4, 15, 20, 21] spends more tokens thinking before committing to an answer. *Sampling scaling* [1, 19] draws more independent attempts and selects one. In both, every additional token is a function of the same frozen weights and the same fixed prompt: more compute reorganizes information already in the system but imports none.

A third axis breaks this closure: *interaction* with an external environment that returns **grounded** feedback—a test result, a rendered layout, a search verdict. Such feedback is not a function of the weights; it is an observation of the artifact's real behavior, and it can carry information the model never had. This axis is well established empirically [6, 13, 14, 22], but accounts of *why* and *when* it should beat reasoning- or sampling-only scaling remain informal, and—we argue—half-stated.

Grounding is the variable, on both sides of the loop. Our thesis is that one variable governs interaction scaling: **grounding**, the property that a signal is computed from the artifact's actual behavior rather than from a model's opinion of it. And grounding must hold on *both* sides of the proposer–reviewer loop (Figure 1):

1. **Grounded feedback** (the reviewer signal that drives revision): run the code and read `pytest`; render the HTML and measure the real bounding boxes; issue the query and read the result. This is the side prior work emphasizes, and a data-processing-inequality argument (Section 3) shows it is exactly what lets interaction exceed the ceiling that bounds reasoning and sampling.
2. **Grounded evaluation** (the metric that *measures* the gain): the same observation, used as the score. This side is almost always overlooked, and getting it wrong silently hides the effect.

The second point is not a detail. The default evaluator for visual artifacts is a vision–language model (VLM) reading a screenshot, and it is *structurally* blind to the defects that matter: content overflowing the canvas is cropped off-frame before the image reaches the judge, and small overlaps fall below its acuity. On dense academic figures this VLM judge rates 14 of 15 single-shot renders "perfect" (an initial 15-figure probe) when a deterministic check finds only 3 are actually clean (Section 7). Worse, when the same VLM is used as the *reviewer*, its edits make slide geometry *worse*. Ungrounded on either side, the third axis either does not fire or cannot be seen.

Contributions.

1. **A grounding framework (Section 3).** A Type 0–3 feedback taxonomy and a data-processing-inequality argument that motivates why ungrounded review and reasoning (and best-of-N selection) saturate while grounded feedback need not—plus the symmetric observation that ungrounded *evaluation* cannot measure the gain. Beyond the documented *biases* of model-as-judge evaluation [23], we isolate a sharper failure: for artifacts whose quality is a measurable physical property, a screenshot judge is *structurally blind* (not merely noisy), which silently hides a real interaction-scaling effect.
2. **Interaction is a distinct, unbounded axis (Sections 5 and 6).** At a matched 20K-token budget, reasoning and best-of-N saturate (73.3%/86.7%) while every interaction strategy reaches $\geq$ 97.8%; an in-modality control isolates test *execution* as the active ingredient, and the harness is the most token-efficient, zero-variance strategy.
3. **Grounded evaluation, and a deterministic instrument (Section 7).** We show the VLM judge is structurally blind, build a deterministic DOM-geometry+alignment instrument, and report large, significant, CI-backed defect reductions under grounded feedback across four visual modalities measured identically—gains the standard metric cannot even see—plus a controlled ablation in which VLM feedback makes layout *worse*.

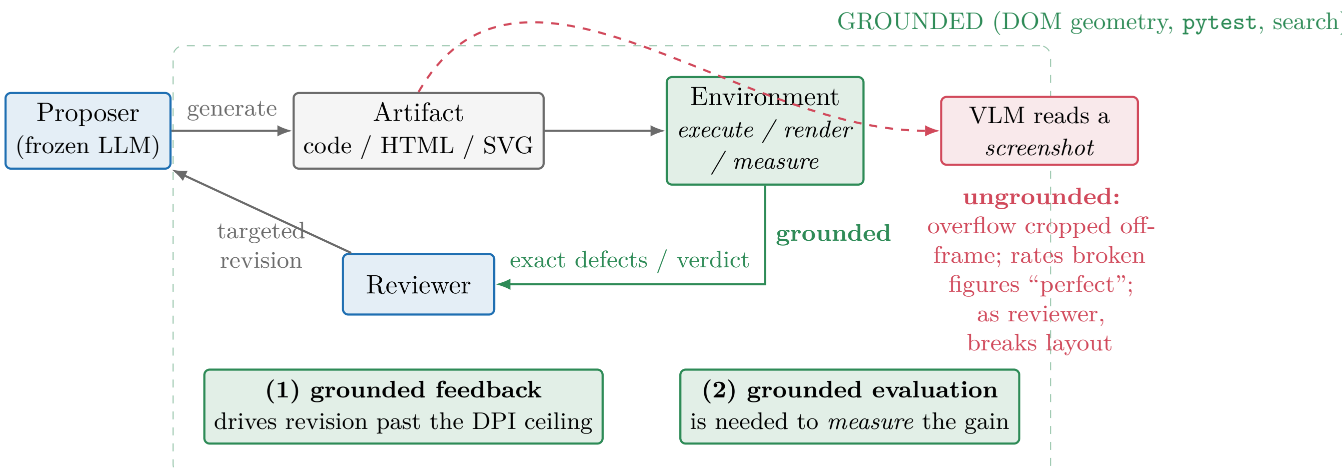

Figure 1: **Grounding must hold on both sides of the interaction loop.** A proposer emits an artifact; an environment *executes/renders/measures* it and returns a grounded signal that the reviewer turns into a targeted revision. Grounded *feedback* (1) is what lets interaction import information from outside the weights and exceed the reasoning/sampling ceiling (Section 3). Grounded *evaluation* (2) is what makes the gain measurable: the default VLM-on-a-screenshot evaluator (red) is structurally blind to off-canvas overflow and fine overlap, so it neither measures nor (as a reviewer) fixes the defects.

4. **Robustness and internalization (Sections 8 and 9).** The effects replicate across model families (including grounded geometry on Gemini 3.1 Pro), to held-out tasks, and under a propose-heavy budget allocation; and the harness's behavior partially distills into an 8B student at $\sim$10$\times$ lower deployment cost.

Availability. All code, task suites, prompts, the deterministic instruments, and an interactive results browser with per-task single-shot-vs-reviewed renders are available at `https://github.com/bojieli/interaction-scaling-paper`.

2 Related Work

Test-time compute scaling. Two axes are established. *Reasoning* scaling—chain-of-thought, tree search, and extended thinking [4, 15, 16, 20, 21]—spends more tokens before committing. *Sampling* scaling—self-consistency and best-of-N [1, 19]—draws multiple attempts and selects. Both transform the same frozen weights and prompt; we make precise (Section 3) the sense in which they are bounded, and position *interaction* as a third axis that imports external information.

Self-correction and critique. Iterative refinement with model-generated feedback—Reflexion [14], ReAct [22], Self-Refine [9], CRITIC [6]—is widely used, but its reliability is contested: Huang et al. [7] show models often cannot self-correct reasoning without an external signal, and Kamoi et al. [8] chart when correction does and does not help. Our taxonomy makes the distinction explicit: *ungrounded* critique (Type 1) is bounded, whereas refinement against an executed/measured result (Type 3) is not—and our in-modality control (Section 6) shows it is the external execution, not the extra critique pass, that carries the gain.

Grounded and execution feedback. Tool use and execution feedback—Toolformer [12], Gorilla [10], Voyager [18], RLEF [5]—and the thinking-vs-doing analysis of Shen et al. [13] establish that acting on an environment helps. We contribute the information-theoretic account of *why*, a feedback-actionability taxonomy that predicts *when*, and—crucially—the requirement that the *evaluation* be grounded too. We also compare harness architectures (single-agent loops [17], sample-and-select [11], proposer–reviewer) under a matched budget.

Model-as-judge reliability. LLM- and VLM-as-judge evaluation is now standard [23], and its failure modes—position, verbosity, and self-preference biases, and broader reward-model fragility [2]—are documented. Our finding is sharper and, to our knowledge, not previously isolated: for artifacts whose quality is a measurable *physical* property (the geometry of a rendered layout), a screenshot judge is not merely biased but *structurally blind*—content that overflows the canvas is cropped off-frame before it reaches the judge—so it both fails to score the defect and, used as a reviewer, fails to fix it. This is why a deterministic instrument is necessary, not merely preferable, to detect interaction scaling on visual modalities.

Information theory. The data-processing inequality [3] is the lens for our bound: any post-processing of the model's outputs (reasoning, ungrounded review) cannot increase information about the target, whereas an environment observation introduces a new channel.

3 The Grounded-Feedback Framework

Our contribution is not a new self-correction method but an account of *why* interaction is a separate axis, *when* it helps (a feedback-actionability taxonomy), and a previously-missing requirement—that the *evaluation* be grounded too, without which the effect is unmeasurable.

3.1 A feedback taxonomy by grounding

We classify the reviewer's feedback channel by how much external, task-conditional information it injects:

- **Type 0 – none.** Single-shot; no feedback.
- **Type 1 – ungrounded critique.** An LLM/VLM critiques the artifact from its weights ("looks wrong"). No execution, no measurement.
- **Type 2 – static analysis.** Linters/type-checkers: signal about the artifact's *form*, not its behavior.
- **Type 3 – grounded.** The artifact is exercised and the result observed: **3a** execution (run tests, read tracebacks), **3b** rendered geometry (measure the DOM), **3c** temporal (sample animation frames), **3d** factual (search and verify claims).

3.2 Why grounding breaks the ceiling: a DPI bound

Let $A^\star$ be the (task-determined) correct artifact and A a candidate. The proposer is a channel from its inputs Θ (weights) and prompt X to A. A *reviewer* that only reads A and reasons forms the Markov chain $A^\star \to (\Theta, X) \to A \to R$, where R is its critique. By the data-processing inequality,

$$I(A^\star; R) \;\le\; I(A^\star; (\Theta, X)),$$

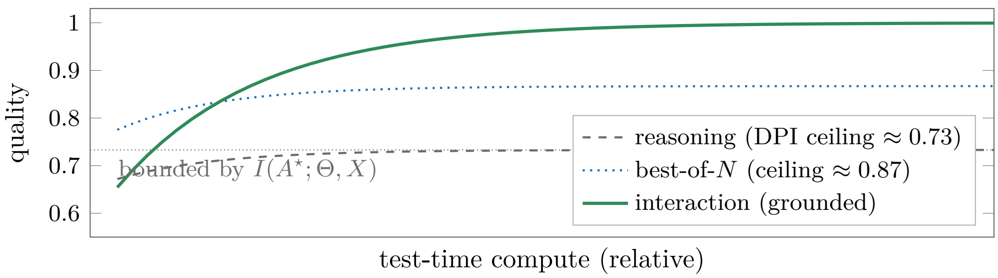

Figure 2: **The DPI ceiling.** Reasoning and best-of-N are functions of the frozen weights/prompt (Θ, X) and saturate at the information already present (measured 73.3% and 86.7% on hard code at a 20K-token budget; Section 5). Grounded interaction imports external information each cycle and exceeds the ceiling, reaching strict 100%. Schematic; saturation values are the measured ceilings.

so *no* amount of reasoning or ungrounded review (Type 0–2) can recover more information about $A^\star$ than is already in (Θ, X). This bounds reasoning-only scaling. Sampling is bounded differently but no less tightly: best-of-N can only *select* among the N candidates the proposer draws, so even an oracle (grounded) verifier cannot return a correct artifact that was never sampled—its ceiling is the support of the proposer's own output distribution, not a new information channel. Both ceilings appear empirically (Figure 2): on hard code, reasoning saturates at 73.3% and best-of-N (with an oracle verifier) at 86.7%. We stress that the inequality bounds the *information channel*, not achievable quality directly; we use it to *motivate* the saturation we then measure, and treat that measured saturation as the evidence.

Grounded feedback (Type 3) escapes both ceilings by feeding an environment observation $E = g(A, \text{world})$—a test outcome, a measured bounding box, a search verdict—back into *generation*. Conditioning the next proposal on E lets the proposer synthesize a candidate it would not otherwise have sampled: formally $I(A^\star; (R, E))$ can exceed $I(A^\star; (\Theta, X))$ because E carries information obtained by *running* the artifact against reality. This is the crucial difference from selection (best-of-N cannot create a missing candidate) and from reasoning (which adds no information), and it is why interaction is a distinct axis rather than a constant-factor speedup.

The symmetric claim: ungrounded evaluation cannot measure the gain. The same inequality applies to the *scorer*. A metric M that is itself a function of (Θ, X)—e.g. a VLM judging a screenshot, whose view of A is a lossy, cropped image—satisfies $I(A^\star; M) \leq I(A^\star; (\Theta, X))$ and cannot certify an improvement that lives in the part of A it cannot observe. Grounding the evaluation (measure the artifact directly) is therefore not optional polish: it is a precondition for *detecting* interaction scaling on any modality whose quality is not fully visible to the default judge. Section 7 shows this is exactly the situation for rendered visual artifacts.

3.3 Predictions

The framework makes three testable predictions, each tested below: **(P1)** reasoning- and sampling-only scaling saturate strictly below the interaction ceiling (Section 5); **(P2)** within interaction, more actionable grounded channels lift more, and the *execution* of the artifact—not the extra review pass—is the active ingredient (Section 6); **(P3)** on modalities whose quality is invisible to the default judge, the gain is only detectable with a grounded metric, and an ungrounded reviewer can even regress quality (Section 7).

Table 1: Modality instantiations. Each modality pairs a grounded feedback channel with the instrument that both drives revision and scores the result. "†" marks the four modalities scored by the deterministic DOM-geometry instrument and analyzed jointly in Section 7.

Modality	Type	Grounded instrument	Hard suite (N)
Code	3a	`pytest` pass/fail + traceback	15 (+11 deep-spec)
Academic figures†	3b	DOM geometry + alignment	20
Slides†	3b	DOM geometry + alignment	12 real-paper
Web pages†	3b	DOM geometry, multi-width	20
Animations†	3c	DOM geometry, per-frame	20 SVG/CSS
Video editing	3a+3c	script exec + Gemini-native rubric	15
Deep research	3d	search-grounded factual rubric	15

4 Method: A Budget-Aware Proposer–Reviewer Harness

Architecture. The harness is pure scaffolding around a *frozen* frontier model (Claude Sonnet 4, `claude-sonnet-4-20250514`, temperature 0 unless noted): no fine-tuning, no retrieval, no learned controller. A **proposer** emits an artifact; an **environment** executes/renders/measures it to produce a grounded signal; a **reviewer**—the same model in a diagnostic role—converts that signal into a typed defect list; the proposer revises. The loop repeats up to an iteration cap and *keeps the best-scoring iteration* (so the reviewed condition can never score below single-shot under the same metric). Single-shot is one proposer call.

The three grounded instruments. Every modality is paired with a deterministic or near-deterministic instrument that both *feeds back* and *scores*:

- **Execution** (`pytest`): exact pass/fail plus the failing assertion and traceback.
- **DOM geometry**: the artifact is rendered headless and every element's true bounding box is read from the layout engine; we compute, exactly, text-on-text overlap ($\geq$ 6 px), out-of-bounds/clipping, container overflow, document overflow ($>$ 16 px), and—new here—*box-group misalignment* (rows/columns of card or pillar boxes flagged for unequal size, misaligned far edges, or uneven gutters). For scrollable web we score horizontal overflow at desktop (1920) and mobile (375) widths; for animations we probe the DOM at each sampled frame.
- **Native video** (Gemini 3.1 Pro): the rendered clip is judged whole against per-requirement binary checks—a near-grounded substitute where pixels, not a still, are observed.

Where no deterministic instrument exists (flowing web content, factual research), we fall back to a binary per-requirement rubric, flagged as such.

Budget. A run has a token budget B split across proposer and reviewer calls; Section 8 sweeps the split. Table 1 lists the modality instantiations.

All artifacts are produced under a common *design-principle* generation prompt (proximity, alignment, repetition, contrast, deliberate color) so that single-shot quality is already strong; this makes the headroom we measure a property of genuine task difficulty rather than a weak prompt.

Evaluation protocol. Unless noted, the proposer runs at temperature 0 and each modality is evaluated over three independent on-policy seeds; the reviewed condition keeps the best-scoring iteration (cap ≤ 3 for geometry, ≤ 5 for code), so it can never score below single-shot under the

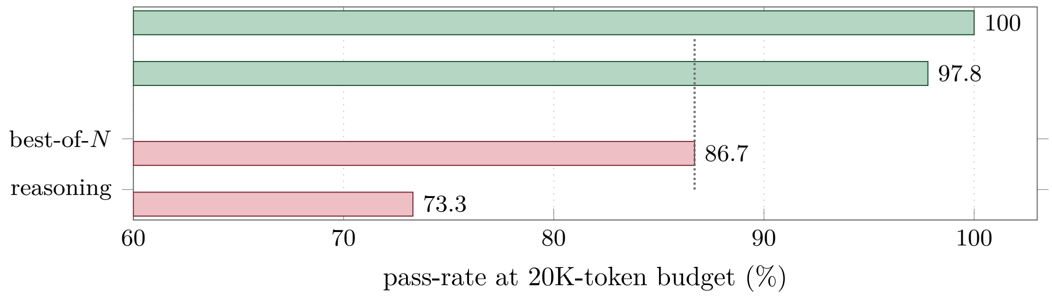

Figure 3: **Two axes saturate; the third does not.** At a matched 20K-token budget on hard code, the two in-token-space axes (red) plateau below the interaction strategies (green), which all reach $\geq$ 97.8%. The proposer–reviewer harness reaches strict 100% across seeds. Dotted line: best-of-N ceiling.

same metric. The unit of comparison is the (task, seed) pair. For each result we report a two-sided paired sign test over decisive pairs and, for the geometric and code effects, a paired-bootstrap 95% confidence interval (10k resamples) on the effect size.

5 Interaction Is a Distinct, Unbounded Axis

Reasoning and sampling saturate; interaction does not (P1). On 15 hard code tasks at a fixed 20K-token budget, we compare four strategies: reasoning-only (extended thinking), best-of-N sampling with an oracle verifier, a single-agent loop (L), and the proposer–reviewer harness (H). Reasoning saturates at 73.3% (86.7% under a generous thinking-heavy budget partition) and best-of-N at 86.7%, while *every* feedback-iterating strategy reaches $\geq$ 97.8% (Figure 3). The gap between interaction and the in-token-space axes is 13–27 pp and is the framework's load-bearing prediction.

Architecture buys efficiency and reliability. Among interaction strategies the gap is not in ceiling pass-rate (they tie within seed noise at N=15) but in cost and variance. The proposer–reviewer harness is the most token-efficient (1,029 mean output tokens vs. 1,431 for the single-agent loop and 1,416 for the oracle-verifier sample-and-select baseline) and the only variant at strict 100% with *zero* seed variance; the single-agent loop hits 100% in 2 of 3 seeds. Context isolation (the reviewer sees only what it needs to localize defects), structured critique (a typed defect list, not raw stderr), and role specialization explain the efficiency.

Built-in early stopping. The loop does not over-revise: on code, across three runs, *every* already-passing single-shot task is submitted at iteration 1 with no revision (30/30 instances, zero wasted tokens). The grounded signal—a failing test—is the trigger; without it the loop is silent. This is why code's token overhead is only ∼37% above single-shot despite an iteration cap of 5.

6 The Active Ingredient Is Execution, Not Review

Code as a clean case study. Code is the cleanest demonstration because the channel is binary and the score is objective. Single-shot pass-rate is 66.7% (10/15); the harness lifts it to a strict 100% in all three runs, recovering every first-shot failure with zero regressions. The mechanism is consistent across recovered traces: turn 1 emits code that compiles but mishandles an edge case (off-by-one boundaries, semantic-version comparison, CIDR arithmetic); turn 2 receives the `pytest` traceback—the failing assertion, the offending input, expected vs. actual—and rewrites the localized

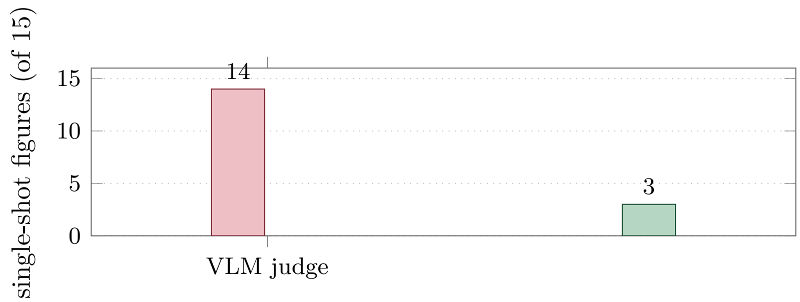

Figure 4: **The VLM judge is structurally blind.** Number of the same 15 single-shot academic figures rated *clean/perfect*: the VLM-on-a-screenshot judge passes 14; the deterministic DOM-geometry instrument passes only 3. The other 11 are broken in ways cropped off-frame or below VLM acuity.

block; turn 3 confirms. Reasoning alone cannot supply this: the model has no way to know its boundary condition is wrong until a concrete test exercises it. A second, harder 11-task suite of deep-spec implementations (a from-scratch JSON parser, an A1 spreadsheet-reference resolver, a minimal edit-script diff, an arithmetic/boolean expression evaluator, . . . ; each validated against a reference solution that passes its own tests) corroborates this at larger headroom: single-shot 69.7% $\rightarrow$ strict 100%, +30.3 pp (bootstrap 95% CI $[15.2, 45.5]$ pp), all 10 failing task-runs recovered, 0 regressions (sign-test $p = 0.002$).

An in-modality control isolates execution (P2). The cross-modal ordering could be confounded by task difficulty, so we fix the task suite (the 15 hard code tasks) and vary *only* the feedback channel: a Type 1 reviewer (LLM cross-review, no execution), a Type 2 reviewer (LLM + `ruff` static analysis, still no execution), and the Type 3a baseline. Forbidden from seeing test results, the Type 1/2 reviewers reach a reviewed ceiling of 13/15; the Type 3a (execution) reviewer reaches 14/15 and does so $\sim$2.5$\times$ more token-efficiently (1.53 vs. 2.07 iterations). The separation is not the extra reviewing pass—both have one—but the *execution* of the tests. (Type 1 and Type 2 do not separate because the seeded bugs are runtime-logic errors that `ruff` cannot see; a `mypy`/`semgrep`-catchable bug set is where the strict Type 2 > Type 1 prediction should be tested.)

7 Grounding the Evaluation: The Gain Is Invisible to a VLM

The visual modalities expose the symmetric half of the framework (P3): even when grounded feedback *does* fix the artifact, an ungrounded metric cannot see it.

The default VLM judge is structurally blind. Our starting evaluator is a binary per-requirement rubric scored by a VLM over a screenshot. On 15 dense academic-paper architecture figures, even with native-resolution quadrant tiling, this judge rates 14 **of** 15 single-shot renders "perfect." Yet by direct inspection they are broken—superimposed section titles, labels spilling out of boxes. The reason is mechanical: screenshots are captured at 1080 px, so any content overflowing the canvas is cropped *off-frame before the image reaches the judge*, and small in-frame overlaps fall below its acuity. A deterministic DOM-geometry check finds only 3 **of** 15 are actually clean (mean 1.6 defects). The VLM is blind to precisely the defects that matter (Figure 4).

Table 2: Deterministic geometric-feedback results, one identical configuration across four visual modalities (Sonnet 4, T=0, 3 seeds, alignment-inclusive reward). Δ is the mean defect reduction; CI is a paired bootstrap 95% interval on Δ% (10k resamples); p is a two-sided paired sign test. Web is summed over 1920/375 widths; animations over sampled frames.

Modality	n	SS	Rev.	Δ	95% CI	impr./regr. (p)
Academic figures (20)	60	0.57	0.15	−74%	$[52, 89]$%	17/2 (7×10^{-4})
Dense slides (12)	36	1.25	0.33	−73%	$[45, 93]$%	13/1 (1.8×10^{-3})
Web pages (20)	60	16.1	8.5	−47%	$[30, 62]$%	33/2 (4×10^{-8})
Animations (20)	60	16.9	10.2	−40%	$[10, 64]$%	34/7 (3×10^{-5})

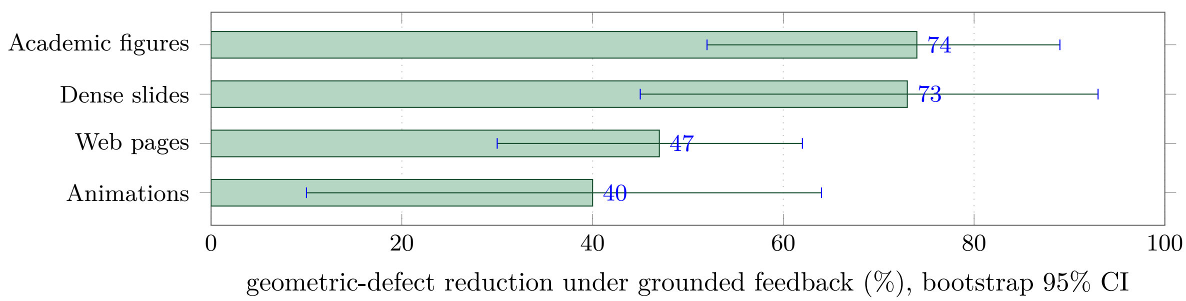

Figure 5: **Grounded geometric feedback removes real layout defects on all four visual modalities** (mean reduction; whiskers are paired-bootstrap 95% CIs, all excluding zero). The standard VLM-on-a-screenshot metric measures none of this.

A deterministic instrument reveals a large, significant effect across four modalities. We score four visual modalities with the same DOM-geometry+alignment instrument under one identical configuration (Sonnet 4, temperature 0, design-principle prompt, three seeds, propose→measure→feed-exact-defects-back→revise, $\leq$ 3 iterations). Table 2 and Figure 5 report the result: grounded geometric feedback produces a large, significant reduction in real layout defects on *all four*, with bootstrap 95% CIs that exclude zero and improvements outnumbering regressions $\geq$ 5:1. The magnitudes track single-shot headroom—dense responsive web pages are defect-rich out of the box (16 defects/page across desktop and mobile), while a frontier model under a design-principle prompt already lays out many figures and slides cleanly. Animations have the widest interval (a 40% reduction, CI $[10, 64]$%): their per-task defect counts are heavy-tailed across the dynamic suite, so we read the sign test (34/7, $p = 3 \times 10^{-5}$) as the robust statement and the mean as indicative.

An ungrounded *reviewer* makes layout worse (ablation, two modalities). Grounding matters on the feedback side too, controllably. Holding tasks and proposer fixed and varying *only* the reviewer, a VLM-feedback reviewer *increases* mean geometric defects on both modalities tested—slides 1.89 → 2.44 and figures 0.52 → 0.62 (regressions outnumber improvements 10:5 on figures)—because, blind to the true geometry, its edits break alignment as often as they fix it. The deterministic geometric-feedback reviewer instead reduces defects on the same suites (1.25 → 0.33 on slides, 0.57 → 0.15 on figures). The binary VLM rubric on these slides simultaneously saturates (0.97 mean, 29/36 rated perfect), confirming the judge cannot even see the difference. Both halves of Figure 1 are necessary (Figure 6).

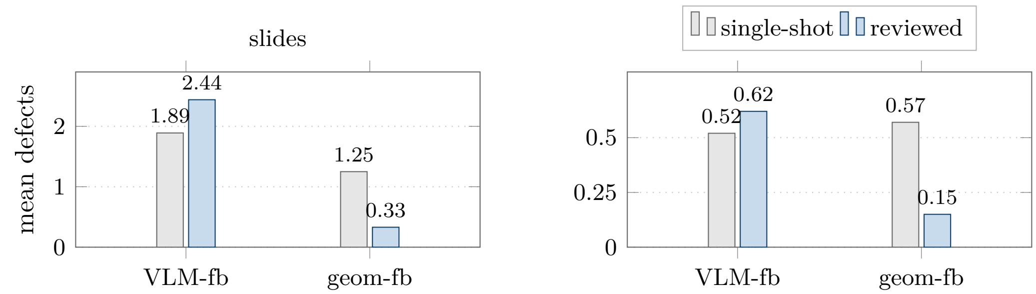

Figure 6: **An ungrounded reviewer makes layout worse (controlled ablation, two modalities).** Same tasks and proposer; only the reviewer's grounding varies. A VLM-feedback reviewer (VLM-fb) *increases* mean geometric defects on both slides (1.89 → 2.44) and figures (0.52 → 0.62); the deterministic geometric-feedback reviewer (geom-fb) reduces them (1.25 → 0.33; 0.57 → 0.15). The grounded signal is necessary on the feedback side, not just the metric side.

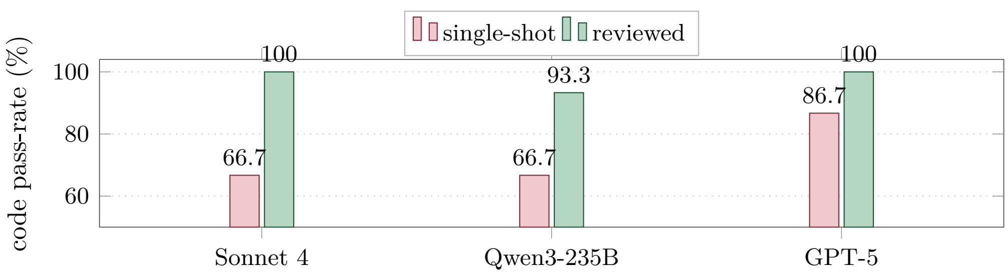

Figure 7: **The execution-grounded loop helps every model family.** Single-shot vs. harness-reviewed code pass-rate for three model families (+33.3/+26.7/+13.3 pp).

Two modalities where the honest answer is "no headroom." *Video editing*, scored by Gemini 3.1 Pro's native full-clip rubric (a near-grounded instrument that observes pixels, not a still), is already strong single-shot (0.70, 22/45 task-runs perfect) so the reviewed lift is small and not significant (+0.04); a hardened multi-step suite de-saturates it (single-shot "fully correct" 49% → 11%). *Deep research*, scored against exact provided facts, saturates (0.982, +0.018, n.s.): a frontier model knows well-documented facts and does not take the planted traps, so the Type-3d lift is bounded by the near-zero single-shot error rate—a genuine limit of the factual modality, since de-saturating it would require facts the judge itself cannot grade. We report these as scoped negatives rather than headline lifts.

8 Robustness

Cross-model, code. The effect is architectural, not a Claude artifact. On the 15 hard code tasks the harness lifts Sonnet 4 by +33.3 pp (to 100%), Qwen3-235B by +26.7 pp, and GPT-5 by +13.3 pp (on a higher single-shot baseline; Figure 7). All three families benefit from the same execution-grounded loop.

Cross-model, grounded geometry. To rule out a Claude-specific or prompt-specific explanation of the grounded-evaluation effect, we run the same deterministic geometric-feedback harness with **Gemini 3.1 Pro** as the proposer on the dense real-paper slide suite, scored by the identical model-free DOM instrument. The effect reproduces and is in fact larger: mean defects fall 3.50 → 0.25 (−93%) over three seeds (n=36), geometrically-clean slides rise 17/36 → 34/36, and 19 of 20 decisive

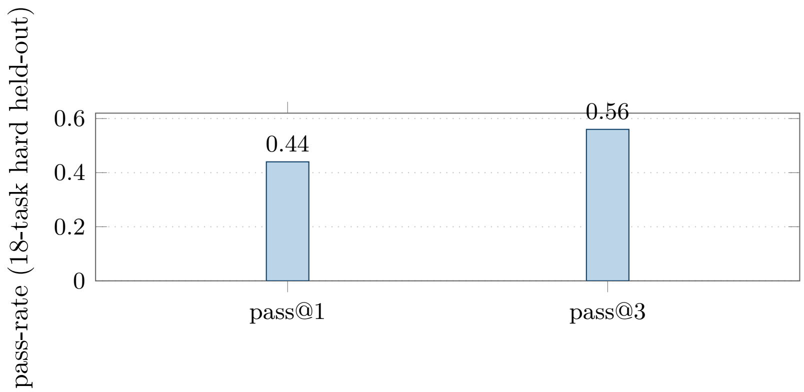

Figure 8: **An 8B student internalizes interaction quality.** Pass@1/pass@3 of the distilled 8B markdown student on the hard held-out suite (0.50×/0.70× teacher mean@1/@2 on the 44-task OOD set), at ∼10× lower deployment cost.

task-runs improve (sign-test $p = 4 \times 10^{-5}$). Gemini's single-shot slides are *more* defect-prone than Sonnet's (3.50 vs. 1.25), and the grounded loop removes almost all of it. Because the instrument depends on no model, this rules out a Claude-specific or scorer-specific explanation.

Held-out generalization. On a 32-task held-out code suite constructed after the method was fixed, the harness lifts 90.6% → 100%, recovering 3/3 first-shot failures with zero regressions—no tuning to the original tasks.

Budget allocation. Sweeping the proposer/reviewer token split on a fixed budget produces an 86.6 pp spread in pass-rate, monotone in the proposer's per-call share: the optimum is *propose-heavy* (allocate the majority to generation, reserve enough for the reviewer to localize defects). Extra inference compute is better spent on additional samples than on deeper iteration past the early-stop point.

9 Internalizing the Harness into a Small Student

Can the harness's interaction-quality behavior be *internalized* so a small model emits high-quality artifacts in fewer passes? We distill judge-filtered teacher trajectories from the harness into an 8B Qwen3-VL student via SFT.

Result. The student reaches 0.50× the teacher's mean@1 capability on a 44-task out-of-distribution suite (rising to 0.70× at pass@2), and 44%/56% pass@1/pass@3 on an 18-task hard held-out suite, at ∼10× lower deployment cost than running the harness over a frontier model (Figure 8). Run inside the same harness, the small student recovers more of the gap, confirming that scaffolding and internalization compose.

Variance is a budget (a cautionary finding). Reinforcement fine-tuning (RFT) on top of SFT improves every per-turn consistency metric we measure *and* regresses pass@3 by 17 pp. For sampling-based deployment, output-distribution variance *is* the inference-scaling lever, and any post-training that reduces it spends from the same budget best-of-N draws on. Variance is not noise to minimize; it is the resource sampling converts into quality. The operational recipe is therefore:

SFT to internalize interaction quality, keep the proposer's sampling temperature, and run it inside the grounded harness.

10 Discussion and Conclusion

Grounding is the load-bearing variable. Two of the three test-time axes are bounded by the data-processing inequality because they never leave the model's token space; the third is unbounded precisely because grounded feedback imports external information each cycle. We have argued and shown that this grounding must hold on *both* sides of the loop. On the feedback side, an in-modality control isolates test *execution*—not the extra review pass—as the active ingredient, and a VLM *reviewer* of slide layout actively regresses quality. On the evaluation side, the default VLM-on-a-screenshot metric is structurally blind to the geometric defects the loop fixes, so the entire visual-modality effect is invisible until the metric is grounded in deterministic DOM geometry—at which point grounded feedback removes 40–74% of real layout defects across four modalities, every CI excluding zero.

A methodological warning for the field. Much of the recent excitement about "LLMs that critique and improve their own visual output" is measured with VLM judges. Our results imply many such measurements are uninformative on layout: the judge cannot see the defects, and using it as the reviewer can make things worse. Where an artifact's quality is a measurable physical property—geometry, execution, retrieval—the evaluation should be grounded in that property, not in a model's picture of it.

Limits, honestly stated. Two modalities saturate: video editing is already strong single-shot (the lift is real only on a hardened multi-step suite), and deep research saturates because frontier models know well-documented facts and the judge cannot grade facts they do not. Deterministic geometry applies to artifacts with a measurable intended layout (figures, slides, web, animations); for genuinely flowing content we still rely on a binary rubric, with the VLM-reliability caveat.

Future work and broader impact. Two directions follow directly. First, deterministic instruments for modalities beyond layout and execution—differentiable or symbolic checks for audio, 3D, tabular, and UI-interaction artifacts—would extend grounded evaluation to where VLM/LLM judges are likewise blind. Second, our DPI argument bounds an information channel; tightening it into a quantitative relationship between feedback fidelity and achievable quality is open. On impact: grounding the metric is a cheap, deployable correction to a measurement error that, we argue, currently inflates reported progress on self-improving visual generation; the same instruments are auditable and reduce reliance on opaque model judges. The harness itself is scaffolding over a frozen model, so it inherits that model's capabilities and failure modes rather than introducing new ones.

Operational recipe. Wrap a frozen frontier model in a proposer–reviewer harness with the most actionable grounded feedback channel for the modality. For visual artifacts, ground *both* the reviewer and the scorer in deterministic geometry—measure the rendered DOM, never a screenshot. Allocate the majority of the per-task budget to the proposer; spend extra compute on samples, not deeper iteration. For cost-sensitive deployment, distill the proposer into a small student and run it inside the same harness. Interaction scaling is real, distinct from reasoning and sampling, predictable from the feedback taxonomy, and—once you ground the metric—plainly visible.

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A Detailed Results and Configurations

This appendix collects the full tables underlying the summaries in the main text: the four-strategy scaling curves, the matched-budget reasoning baseline, the in-modality feedback-type control, the single-agent-vs-harness comparison, cross-model and held-out code, the budget-allocation simplex, token ROI by modality, the distillation ablations, the per-task pass@k breakdown for the 8B student, and the original six-modality holistic overview (superseded for the visual modalities by the grounded measurements of Section 7).

Table 3: **The headline scaling result (3-seed mean ± SD).** Pass-rate by strategy × budget on 15 hard code tasks, Sonnet 4. S, L, and H are averaged over three independent on-policy runs; R was run once (extended-thinking variation across seeds is small at temp 0) at the sweep's budget partition (a more generous partition lifts R to 86.7%; see footnote and Table 4). S's small dip from B=1K (80.0%) to B=5K (77.8%) is within the ±3.1 pp seed noise on a 3-seed × 15-task suite; pass@N is non-decreasing in N in expectation but not pointwise across finite-sample on-policy seeds. *Reasoning-only and best-of-N saturate well below 100%*; all three iteration-with-feedback strategies reach ≥ 97.8% at B=20K (L hit 100% on 2 of 3 seeds; H on all 3; H is the only variant at strict 100% across seeds). The architectural ordering across H, L is within seed noise on pass-rate; what separates them is token efficiency and reliability (Section 5).

Budget B	R (reasoning-only)	S (best-of-N)	L (single-agent loop)	H (proposer–reviewer)
1K	60.0%	80.0% ±0.0 pp	57.8% ±3.1 pp	62.2% ±3.1 pp
5K	73.3%	77.8% ±3.1 pp	**93.3%** ±0.0 **pp**	91.1% ±3.1 pp
20K	73.3%	86.7% ±0.0 pp	97.8% ±3.1 pp	**100.0%** ±0.0 **pp**
Ceiling	73.3%	86.7%	∼100%	**100% (zero variance)**

Table 4: Reasoning-only at matched budget vs. single-shot and the harness on the 15 hard code tasks. Single-seed numbers (the Phase-1 reference seed, also used in Table 5); the 3-run aggregate is reported in Table 16 (66.7 ± 6.7% single-shot, 100.0 ± 0.0% reviewed). Reasoning closes two-thirds of the harness's gap over single-shot but falls 6.7 pp short of the harness's pass-rate even at 1.57× the harness's token budget.

Strategy	Pass-rate	Mean tokens	Δ vs. single-shot
Single-shot	73.3% (11/15)	1,406	—
Reasoning-only (matched budget)	86.7% (13/15)	4,137	+13.3 pp
Proposer–reviewer harness	**93.3% (14/15)**	**2,643**	+**20.0** pp

Table 5: In-modality feedback-type controls on 15 hard code tasks (Sonnet 4 proposer and reviewer). Each row is a single on-policy seed; the "SS" column is the *per-seed* single-shot pass-rate (not held constant across rows) and "Reviewed" is the ceiling after that seed's harness run. The three SS values (10/15 and 11/15) are draws from the same single-shot distribution that gives Table 16's 3-run mean of $66.7 \pm 6.7\%$ (i.e. within the ± 1 SD band centered on 10/15). Because the SS column varies by row, the load-bearing comparison is the *reviewed* ceiling: Type 3a reaches 14/15, Type 1/2 reach 13/15 (+6.7 pp), *and* Type 3a is $\sim 2.5\times$ more token-efficient (1.53 vs. 2.07 iterations).

Type	Reviewer sees	SS	Reviewed	Iters
None (single-shot)	—	11/15	—	—
Type 1 (cross-review)	code only	10/15	13/15	2.07
Type 2 (+`ruff`)	code + static analysis	11/15	13/15	2.07
Type 3a (+exec)	code + test stderr/traceback	11/15	**14/15**	**1.53**

Table 6: Three interaction-strategy variants at B=20K on the 15 hard code tasks, all using Type 3a execution feedback. Pass-rate is within seed noise (sign-test p>0.6 on the H–L comparison); the harness wins on *token efficiency* (1,029 vs. 1,431 L vs. 1,416 IAD mean output tokens) and on *seed variance* (zero across three seeds for H; L hit 100% in 2 of 3 seeds; IAD was run for one seed only).

Strategy	Pass-rate	Mean tokens	Notes
L (single-agent loop)	97.8% ±3.1 pp (3 seeds)	1,431	1 task in 1 seed
IAD (oracle, K=3, T=0.7)	100.0% (1 seed)	1,416	+38% tokens vs. H
H (proposer–reviewer)	**100.0% ±0.0 pp (3 seeds)**	**1,029**	zero seed variance

A.1 Four-strategy scaling at a matched 20K-token budget

A.2 Reasoning-only at a matched budget

A.3 In-modality feedback-type control (Type 1/2/3a) on a fixed code suite

A.4 Single-agent loop vs. proposer–reviewer harness

A.5 Cross-model replication (code)

A.6 Held-out generalization (code)

A.7 Budget-allocation simplex

A.8 Token ROI by modality

A.9 Distillation: headline

A.10 Distillation: capability U-curve over force-close budget

A.11 Distillation: data-scale

A.12 Distillation: RFT regresses pass@3

A.13 Per-task pass@k for the 8B markdown student

A.14 Original six-modality holistic overview (historical)

Table 7: Cross-model replication on the same 15 hard code tasks. The Claude row is the multi-run reference from Table 16; Qwen and GPT-5 are single on-policy runs at T=0.7. The harness lift survives across three families (Anthropic, Alibaba, OpenAI).

Model	Single-shot pass	Reviewed pass	Δ (lift)
Claude Sonnet 4 (3-run mean)	$66.7 \pm 6.7\%$	$\mathbf{100.0 \pm 0.0}\%$	+**33.3** pp
Qwen3-235B-Instruct-2507 (1 run)	66.7% (10/15)	**93.3**% (14/15)	+**26.7** pp
GPT-5 (1 run)	86.7% (13/15)	**100.0**% (15/15)	+**13.3** pp

Table 8: Held-out generalization on 32 newly constructed code tasks (zero overlap with the 15-task development set). The harness recovers 3/3 single-shot failures and regresses zero passing tasks. The smaller absolute Δ is a baseline-compression effect (single-shot is 90.6% on the held-out set, leaving at most 9.4 pp headroom).

Split	N	Single-shot	Reviewed	Δ	SS-fail fixes / regressions
Development (3-run mean)	15	$66.7 \pm 6.7\%$	$100.0 \pm 0.0\%$	+33.3 pp	5/5 / 0
Held-out v2	32	90.6%	**100.0**%	+**9.4** pp	**3/3** / **0**

Table 9: Budget allocation simplex on 15 hard code tasks, B = 10K total. Pass-rate is monotone in the proposer's per-call cap; review-heavy corners (low b_1) collapse. The 9-point sweep yields an 86.6 pp spread.

Allocation	b_1 (prop)	b_2 (exec)	b_3 (rev)	Pass-rate	Mean tokens
A (propose-heavy)	0.80	0.10	0.10	**93.3% (14/15)**	3,101
G (prop-dominant)	0.50	0.25	0.25	**93.3% (14/15)**	2,697
D (prop+exec)	0.40	0.40	0.20	80.0% (12/15)	4,025
E (prop+review)	0.40	0.20	0.40	80.0% (12/15)	4,388
I (equal)	0.33	0.34	0.33	66.7% (10/15)	4,845
H (review-dominant)	0.25	0.25	0.50	40.0% (6/15)	7,485
F (exec+review)	0.20	0.40	0.40	13.3% (2/15)	8,771
B (execute-heavy)	0.10	0.80	0.10	6.7% (1/15)	9,383
C (review-heavy)	0.10	0.10	0.80	6.7% (1/15)	9,383
			Spread:	**86.6 pp**	

Table 10: Per-task token cost and quality return-on-investment for the harness, averaged across three on-policy runs. "Extra tokens" is reviewed-minus-single-shot per task. Tokens per 0.01 quality is extra tokens divided by (mean reviewed quality − mean single-shot quality)·100. Code dominates by 100×.

Modality	Single-shot tokens	Reviewed tokens	Extra tokens	Tokens / 0.01 quality
Code	3,336	4,575	1,239	**37**
Video	3,808	20,584	16,776	359
Research	14,116	27,793	13,677	2,415
Webpages	14,060	80,148	66,088	5,128
Animations	25,491	89,009	63,518	4,331
Slides	10,274	36,956	26,682	4,447

Table 11: Distilled 8B student on the 18-task hard held-out suite (teacher's pre-curated single-shot failures, so teacher single-shot is 0/18 by construction). The student lifts judge-keep from 0% to 44% at pass@1 and 56% at pass@3. The 0.50× capability ratio against the teacher is established on a separate 44-task out-of-distribution suite reported next, where teacher single-shot is non-degenerate.

Setup	Compute	Judge-keep
1 sample × 5 turns	1×	44% (8/18)
1 sample × 8 turns	1.6×	28% (regressed)
3 samples × 5 turns (pass@3)	3×	**56% (10/18)**

Table 12: Reasoning-cap U-curve on the 18-task hard held-out suite. SFT recipe sweep on Qwen3-VL-8B-Instruct (37 judge-kept teacher traces; same inference protocol). The optimum is at the student's coherence horizon, not the teacher's.

Variant	Reasoning cap (chars)	Judge-keep	Identical-retry
V1 (no reasoning)	0	6%	67%
V4	1000	11%	35%
V2	3000	17%	32%
V3 (chosen)	**1500**	**44%**	10%

Table 13: Three independent attempts to scale the SFT corpus past V3's 37 examples. All regressed. Quality dominates quantity at this scale.

Variant	Composition	n examples	Judge-keep
V3 (chosen)	235B teacher, judge-kept	37	**44%**
V5	V3 + 13 judge-rejected (235B)	50	33%
V6	V3 + 20 judge-kept (30B teacher)	57	33%
RFT v1	Self-rolled, judge-filtered	61	39%

Table 14: RFT polishes consistency at the cost of pass@k. Per-turn metrics improve; per-task variance collapses; pass@3 regresses by 17 pp. For deployment with sampling, the model's variance *is* the inference-scaling budget. *Note on notation.* **mean@1** is the expected pass-rate of a single random sample, computed as the mean accuracy across the three sampling seeds (one T=0 greedy + two T=0.7). This is the natural variance-sensitive metric and differs from the greedy single-seed *pass@1*=44% reported in Table 11, which is just seed 1's accuracy. *pass@2* and *pass@3* are union pass-rates over the first 2 / all 3 seeds.

	Consistency metrics			Sampling-aware judge-keep		
	id-retry ↓	addresses-rev ↑	no-artifact ↓	mean@1	pass@2	pass@3
V3 (chosen)	9%	83%	2	**31%**	**50%**	**56%**
RFT v1	**8%**	**89%**	**1**	28%	39%	39%

Table 15: Per-task judge-keep outcome for the 8B markdown student on the 18-task hard held-out suite. ✓ = judge-kept; · = judge-rejected.

Category	Task	seed 1 (T=0)	seed 2 (T=0.7)	seed 3 (T=0.7)	pass@3
webpages	web_002	✓	·	·	✓
	web_003	✓	·	✓	✓
	web_005	·	·	·	·
	web_008	·	·	·	·
	web_012	·	·	·	·
slides	slide_001	·	·	·	·
	slide_006	·	·	·	·
	slide_007	·	·	·	·
	slide_008	✓	·	·	✓
	slide_010	·	·	·	·
	slide_014	✓	✓	✓	✓
	slide_018	✓	✓	·	✓
	slide_020	·	✓	·	✓
code	code_h001	✓	·	·	✓
	code_h002	✓	✓	·	✓
	code_h003	·	·	·	·
	code_h004	·	·	✓	✓
	code_h005	✓	✓	✓	✓
	Total kept:	8/18	5/18	4/18	**10/18 (56%)**

Table 16: Single-shot vs. reviewed quality across six modalities, Claude Sonnet 4. Means and standard deviations are computed across three independent on-policy runs (per-run means, then aggregated). Code is binary pass-rate; other modalities are continuous in $[0, 1]$. p-values: one-sided paired sign test on per-task means. **Bold** marks rows where the lift clears $p < 0.05$. Code is scored by execution (`pytest`); video by a native full-video rubric (Gemini 3.1 Pro judging the whole rendered clip); the four visual/factual rows use a holistic screenshot judge and serve only as a cross-modal overview. *This table is historical and is superseded for the visual modalities by the grounded measurements of Section 7*: under the deterministic DOM-geometry instrument (Table 2) the figure, slide, web, and animation effects are large and significant, whereas the holistic-judge numbers here cannot see the geometric defects and therefore mis-state them.

Modality	N	Feedback type	Single-shot	Reviewed	Δ (sign-test p)
Code	15	3a	$66.7 \pm 6.7\%$	$\mathbf{100.0 \pm 0.0}\%$	$+\mathbf{33.3}$ pp (0.008)
Video editing	15	3a+3c	0.70 ± 0.03	0.74 ± 0.05	+0.04 (0.64)
Deep research	15	3d	0.63 ± 0.06	0.69 ± 0.06	+0.06 (0.090)
Web pages	15	3b	0.43 ± 0.03	0.56 ± 0.01	+0.13 (0.073)
Animations	15	3c	0.40 ± 0.03	0.55 ± 0.01	+0.15 (0.073)
Slide decks	20	3b	0.73 ± 0.01	0.79 ± 0.01	+0.06 (0.090)